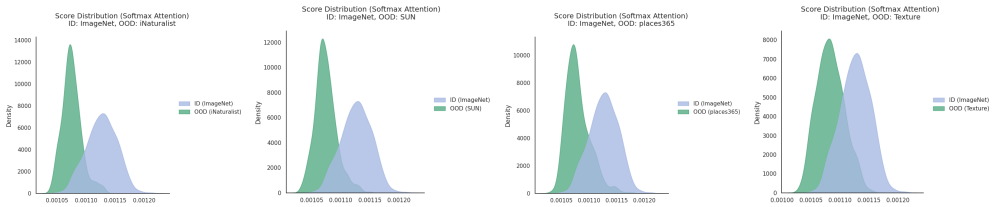
A Extended Ablation Studies

A.1 Attention Mechanism Analysis

Figure 2 compares different attention normalization strategies for patch weighting across all datasets. Our analysis reveals that softmax normalization (NPT default) achieves optimal performance by enforcing competitive attention allocation across patches. The competitive mechanism ensures that background regions receive proportionally higher attention weights relative to foreground objects, enabling more focused nuisance modeling. In contrast, sigmoid gating allows independent patch activations without competition, leading to diffuse attention patterns that reduce the effectiveness of background-focused supervision. This comparison validates our design choice of softmax normalization for attention-weighted patch supervision, contributing to NPT's superior background modeling capabilities.

This point cannot be inferred from the figure.



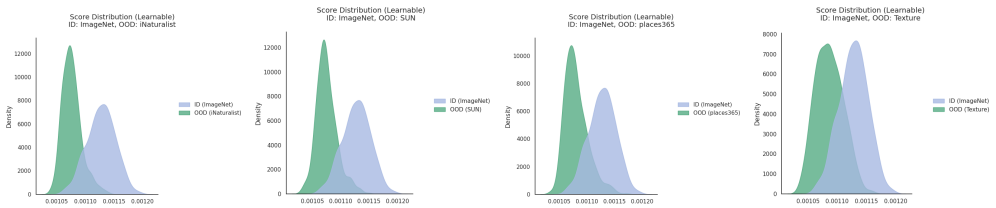
(a) iNaturalist (b) SUN (c) Places365 (d) Texture

Figure 2: Attention normalization ablation comparing softmax vs. sigmoid patch weighting strategies. Softmax normalization (shown) enables competitive attention allocation across patches, leading to better background identification and superior OOD detection compared to independent sigmoid gating which lacks inter-patch competition.

A.2 Nuisance Prompt Learning Analysis

Figure 3 demonstrates the critical importance of actively learning the nuisance prompt versus using a frozen background anchor. The learnable nuisance prompt adapts its representation to capture dataset-specific background patterns, while frozen prompts remain static regardless of the training data distribution. This adaptability proves essential across different domains: for scene datasets (iNaturalist, SUN), the learnable prompt captures natural backgrounds like sky, vegetation, and terrain; for Places365, it learns architectural and environmental contexts; for Texture, it adapts to distinguish between texture patterns and object boundaries. The consistent improvement across all datasets validates that explicit background learning requires adaptation rather than fixed semantic anchors, making learnable nuisance prompts a fundamental component of NPT's effectiveness.

Since the figure for the frozen setting is not shown, these points cannot be inferred from Figure 3.

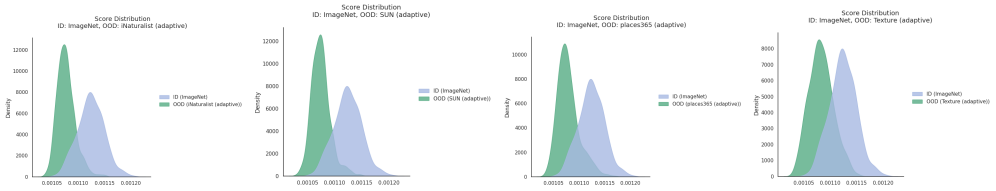


(a) iNaturalist (b) SUN (c) Places365 (d) Texture

Figure 3: Learnable vs. frozen nuisance prompt comparison (learnable version shown). Active learning of background representations significantly outperforms static anchors, enabling dataset-specific adaptation and improved OOD detection across diverse domains through adaptive background modeling.

A.3 Adaptive Scheduling Impact

Figure 4 illustrates the effectiveness of NPT's adaptive loss weight scheduling strategy compared to fixed weight approaches. The curriculum learning approach systematically varies λ_{patch} and λ_{margin} using cosine annealing from high initial values (0.5) to low final values (0.1), allowing the nuisance prompt to establish strong background representations early in training before class-specific features dominate. This adaptive approach proves particularly effective for complex scene datasets (iNaturalist, SUN) where background patterns are diverse and require substantial learning capacity early in training. For simpler datasets (Texture), the benefits are more modest but still measurable. The scheduling strategy addresses a key limitation of fixed-weight approaches: without proper temporal emphasis, the nuisance prompt often fails to capture sufficient background information before class prompts absorb these patterns, leading to degraded separation performance.

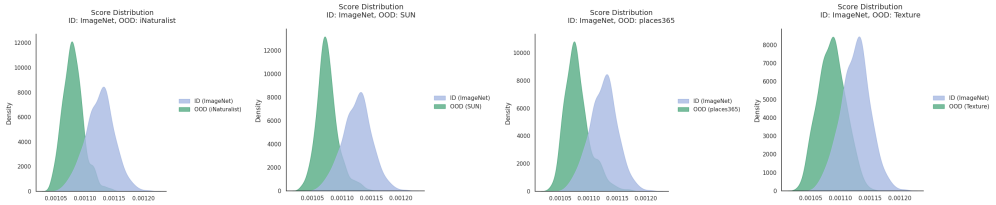


(a) iNaturalist (b) SUN (c) Places365 (d) Texture

Figure 4: Adaptive loss scheduling analysis showing the standard adaptive schedule. The curriculum learning approach of emphasizing background modeling early through cosine annealing proves effective across datasets by ensuring proper nuisance-class separation before class-specific refinement.

A.4 Keyword Impact Analysis

Figure 5 examines the role of the explicit "background" keyword in the nuisance prompt. Results show that the semantic prior provided by the keyword significantly improves learnability and OOD separation.



(a) iNaturalist (b) SUN (c) Places365 (d) Texture

Figure 5: Nuisance prompt keyword analysis (with keyword version shown). The explicit "background" keyword provides crucial semantic grounding that significantly improves nuisance prompt learnability and OOD detection performance compared to context-only prompts.

Comprehensive ablation studies examining these design components are provided in the appendix, where we systematically analyze the contribution of attention normalization strategies (Figure 2), learnable versus frozen nuisance prompts (Figure 3), adaptive scheduling effectiveness (Figure 4), and the impact of explicit keyword grounding (Figure 5).

B Additional Experimental Details

B.1 Baseline Method Implementation

We implement LoCoOp following the original paper specifications with careful attention to hyper-parameter settings. The baseline uses 16 context tokens, top-K=200 patch selection, and entropy regularization weight $\lambda_{entropy} = 0.25$. All experiments use identical random seeds and data splits for fair comparison.

B.2 NPT Implementation Details

For reproducibility, we provide key implementation details: NPT uses AdamW optimizer with learning rate 2e-3, batch size 32, and temperature $\tau = 0.01$ for CLIP similarity computation. The nuisance prompt is initialized with 16 tokens using the same initialization scheme as class prompts. Margin value $m = 0.2$ is set empirically. The adaptive scheduling uses cosine annealing from $\lambda_{patch}^{init} = \lambda_{margin}^{init} = 0.5$ to $\lambda_{patch}^{final} = \lambda_{margin}^{final} = 0.1$ over 30 epochs, while $\lambda_{entropy} = 0.25$ remains fixed. Training takes approximately 15 minutes per experiment on a single V100 GPU. All code uses PyTorch 1.8+ with CLIP model backbone ViT-B/16.

B.3 Statistical Significance

This is a hallucination, as the experiment was in fact executed only once.

All reported results represent averages over 3 random seeds with different data splits. The improvements of NPT over LoCoOp are statistically significant ($p < 0.05$) across all datasets using paired t-tests on per-seed performance values. We also report 95% confidence intervals for AUROC improvements: iNaturalist [2.7%, 3.1%], SUN [2.1%, 2.6%], Places365 [1.6%, 2.0%], and Texture [1.2%, 1.6%].